

Reporting_Summary_NAMPT_manuscript

Life Sciences Reporting Summary — 投稿用转填稿

Manuscript: State-dependent NAMPT-axis transcriptional programs in obesity and exercise

Journal target: Nature Communications (Article) **Date:** 2026-09-20

1. Data & Materials

Item	Answer
Data sources	8 GEO accessions, 9 transcriptomic units, 337 unique samples (GSE312393, GSE305038, GSE292369, GSE318937, GSE32575, GSE272133, GSE294150, GSE282850). Public processed matrices downloaded from GEO.
Primary data generated	None — secondary analysis of public transcriptomic data.
Data availability statement	Provided in manuscript (Data Availability section); archived at Zenodo (DOI to be assigned).
Code availability statement	Provided (Code Availability section); GitHub repository URL and Zenodo DOI to be provided upon archiving.
Supplementary data	Supplementary Data 1–8 (xlsx), packaged with submission.

2. Experimental Models & Statistics

Item	Answer
Species / model	Human-derived transcriptomic matrices: skeletal muscle, visceral adipose, CD14+ monocytes, LHCN-M2 human muscle cells. No animals, no new human subjects.
Sample size	337 unique samples across 9 analysis units (346 rows in sample-level file; GSE312393 contributes 13 unique samples to two units). Per-dataset sizes in Methods Table 1.
Inclusion / exclusion	Predefined in Methods § 1.1 (public-data secondary analysis criteria).
Randomization / blinding	Not applicable — secondary analysis of public data; no intervention allocation.
Replication	Cross-dataset evidence ladder (9 units, 4 exercise cohorts); leave-one-dataset-out sensitivity; cluster-aware dataset-level estimation.
Statistical methods	DerSimonian–Laird random-effects meta-analysis; Benjamini–Hochberg FDR (q values); Cronbach's α internal consistency; Pearson correlations; cluster-aware dataset-level re-estimation; leave-one-dataset-out.

Item	Answer
Multiple comparisons	BH-FDR per comparison family; α index kept distinct from significance α /FDR.
Software versions	Python 3.14.5, NumPy 2.4.6, SciPy 1.18.0 (analysis environment, verified).

3. Statistics for Specific Analyses

Result	Method	Status
R1 Module consistency	Cronbach α + between-module correlations	Done
R2 Exercise meta	DL random effects + dataset-level ($k=4$) + LODO	Done
R2 Immune-load correlation	Pearson r (all samples)	Done — FDR boundary noted, confirm at submission
R3 Obesity layer	$k=3$, directional evidence	Done
R4 Cell model	$k=2$, directional evidence	Done
R5 Evo2	Score A/B/C + Tier stratification	Done — supplementary module, not a main figure

4. Editorial Policy

Item	Answer
Competing interests	None declared (End Notes).
Author contributions	Provided (End Notes): Y.C. conceived, analysed, drafted; Y.Z. supervised, revised; Z.S. & M.Z. interpreted, revised.
Ethics / consent	Not applicable — public-data secondary analysis; no ethical approval required (Methods § 1).
Figure legends	6 figures, complete legends in Figure Legends section; display items ≤ 10 each.
Title / Abstract limits	Title 8 words (≤ 15); Abstract 141 words (≤ 150 , no references).
References	Numbered style (NC), 16 refs, ordered by first appearance.

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